Jiangxi University of Finance and Economics

ArchiMate Layered Architecture and Relational Schema Refinement Report

Luckin Coffee Inc.

A Technology-Driven New Retail Coffee Enterprise

IK2015: Enterprise Architecture and Database Programming

Seminar 6

Group 4

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1 Report Basic Information

Table 1: Basic information of the Seminar 6 report.

Item	Information
Selected enterprise / business	Luckin Coffee Inc.; mobile-first coffee retail, pickup, delivery, partnership-store operation, and packaged coffee business.
Course and seminar	IK2015 Enterprise Architecture and Database Programming; Seminar 6.
Group number	Group 4.
Group members	Zhe-Kai Yang (Krust), Shi-Hang Chen (Sean), Wei-Yu Cai (Nosta), Huan-Wen Chen (Evan), Yuan-Yong Liao (Foreve).
Main deliverable	ArchiMate layered architecture diagrams (Business, Application, Technology) for Luckin Coffee and refined relational schema with normalization documentation.
Connects to	- Seminar 4 — Conceptual ER model (source for relational mapping). - Seminar 5 — Zachman Row 1 Scope and initial relational data model.

This report follows the Seminar 6 instruction sheet. It first presents three ArchiMate layered architecture views — Business Layer Model (BLM), Application Layer Model (ALM), and Technology Layer Model (TLM) — that describe the Luckin Coffee enterprise architecture from stakeholder-facing business processes down to infrastructure. It then refines the relational data model from Seminar 5, documenting normalization to Third Normal Form (3NF), a data dictionary, sample data, and the final relational schema.

2 ArchiMate Layered Architecture Models

Enterprise architecture describes an organization through multiple layers of abstraction. Following the ArchiMate modelling standard, this section presents the Luckin Coffee architecture across three layers: the Business Layer (actors, services, processes), the Application Layer (application services, components, data objects), and the Technology Layer (devices, networks, platform software, and infrastructure artifacts). Together, these three views form a complete architecture description from business motivation to physical deployment.

2.1 Business Layer Model (BLM)

The Business Layer describes the enterprise from the perspective of business actors, roles, services, processes, and objects. It answers the question: *what does the business do, who is involved, and how is value delivered?*

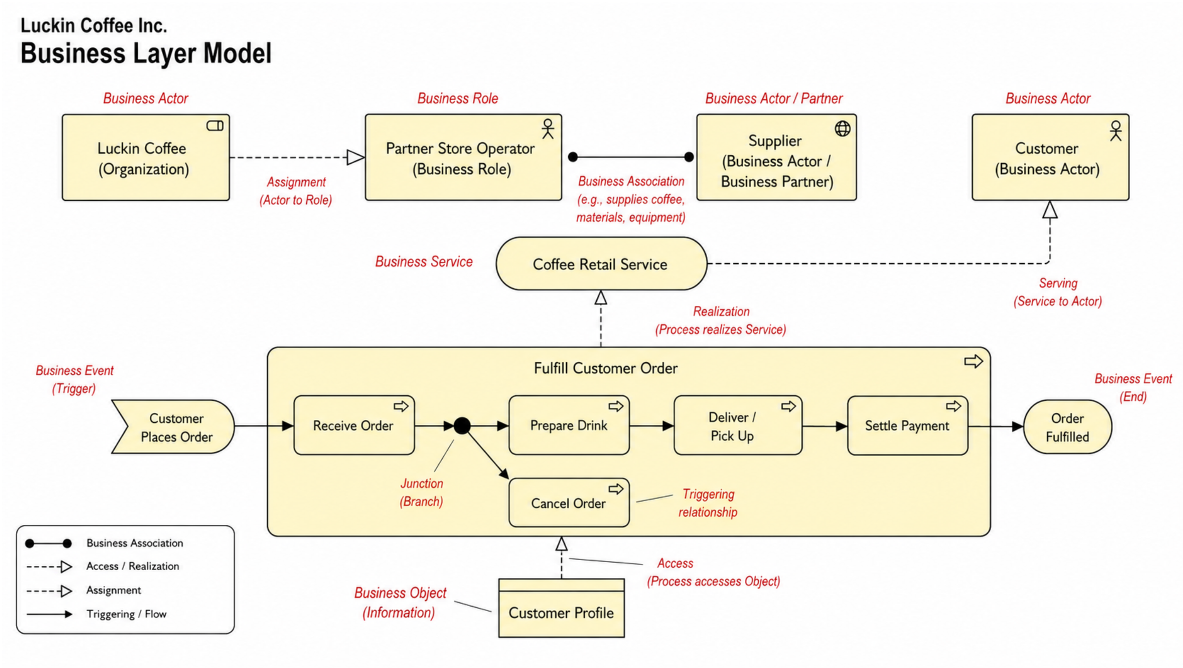


Figure 1: Business Layer Model for Luckin Coffee — showing business actors, roles, the core coffee retail service, and the Fulfill Customer Order business process.

2.1.1 Business Actors and Roles

Table 2: Business actors and roles in the BLM.

Element	ArchiMate Type	Description
Luckin Coffee	Organization	The enterprise itself — the legal entity that owns the brand, platform, and operating model.
Store Operator	Business Role	The operational role responsible for managing store resources, staff scheduling, and physical fulfilment.
Supplier	Business Actor / Partner	External partner providing raw materials (coffee beans, milk, syrup, packaging), equipment, and recipe input.
Customer	Business Actor	The end consumer who places orders, receives beverages, and provides feedback.

2.1.2 Business Service

The model identifies one core business service: **Coffee Retail Service**. This service encapsulates the value Luckin delivers — freshly brewed coffee and tea beverages, fulfilled through pickup or delivery channels, at affordable prices. The Supplier is assigned to support this service, reflecting the integral role of raw-material sourcing in product quality.

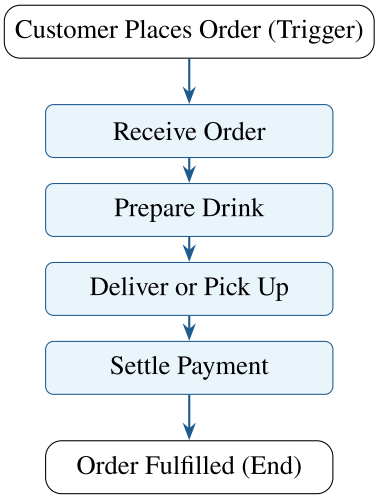
2.1.3 Business Process: Fulfill Customer Order

The central business process is **Fulfill Customer Order**, which decomposes into four sub-processes:

1. **Receive Order** — The customer initiates an order via the mobile app, WeChat mini-program, or other digital channel. The order is captured, validated, and routed to the appropriate store.
2. **Prepare Drink** — Store staff follow standardized recipes to prepare the ordered beverage. Equipment sensors (IoT) may monitor preparation parameters.
3. **Deliver or Pick Up** — The completed order is either handed to the customer for in-store pickup or handed to a delivery rider for last-mile delivery.
4. **Settle Payment** — Payment is processed (pre-paid or on-collection), and the transaction is recorded.

The process is triggered by the business event **Customer Places Order** and terminates with the end event **Order Fulfilled**. The business object **Customer Profile** is accessed by the Customer actor, linking identity and preference data to the service.

2.1.4 Process Flow Diagram



2.2 Application Layer Model (ALM)

The Application Layer describes the software services, application components, and data objects that realize the business processes. It answers the question: *what applications support the business, and how do they manage data?*

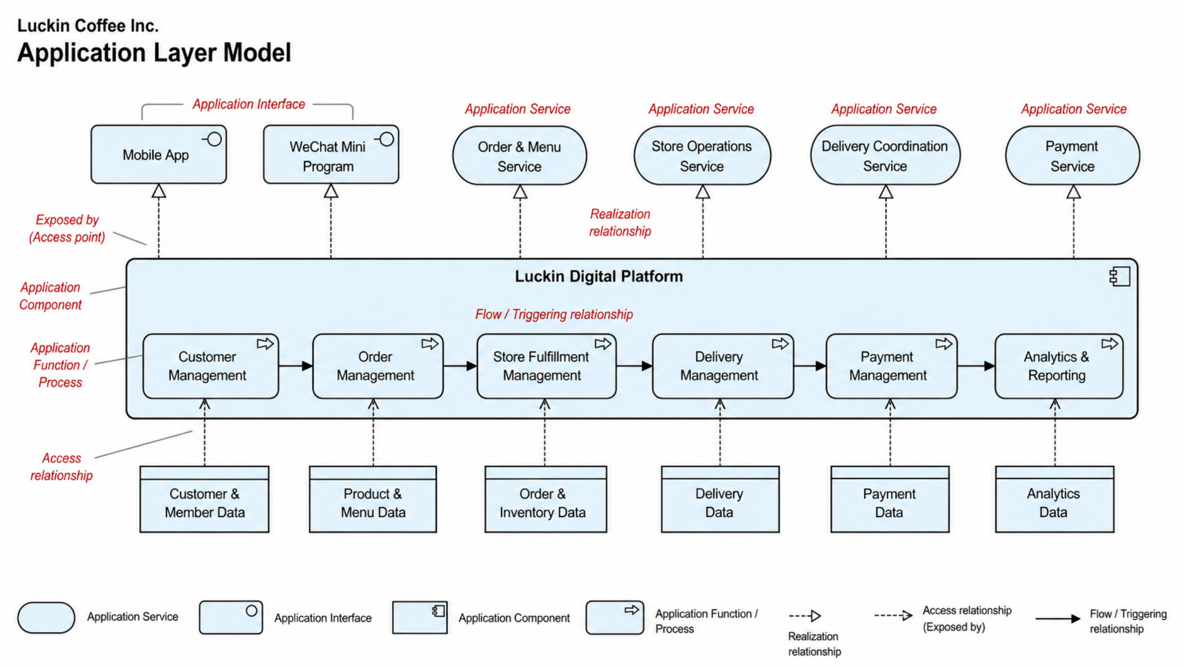


Figure 2: Application Layer Model for Luckin Coffee — showing customer-facing application interfaces, application services, the Luckin Digital Platform components, and managed data objects.

2.2.1 Application Interfaces

In ArchiMate, an **Application Interface** is the mechanism through which a user or external system accesses an Application Service. The ALM identifies two primary customer-facing application interfaces:

Table 3: Application interfaces in the ALM.

Interface	Type	Description
Mobile App	iOS / Android Application	Primary customer-facing interface for browsing the menu, placing orders, managing membership, receiving promotions, and tracking order status.
WeChat Mini Program	Mini Program	Lightweight alternative ordering interface embedded in the WeChat ecosystem, reducing the friction of app installation.

2.2.2 Application Services

The ALM exposes four backend application services, each realizing a distinct business capability:

Table 4: Application services in the ALM.

Service	Type	Description
Order & Menu Service	Application Service	Manages the product catalogue (menu), order creation, order routing to stores, and order status lifecycle.
Store Operations Service	Application Service	Supports store-level functions: queue management, preparation workflow, inventory checks, and staff task assignment.
Delivery Coordination Service	Application Service	Coordinates last-mile delivery: rider assignment, route optimization, estimated delivery time, and delivery status tracking.
Payment Service	Application Service	Processes digital payments across multiple methods (WeChat Pay, Alipay, bank card), records transactions, and handles refunds.

2.2.3 Application Components: Luckin Digital Platform

The platform is decomposed into six internal components, each encapsulating a cohesive set of business logic:

Table 5: Application components within the Luckin Digital Platform.

Component	Responsibilities
Customer Management	Customer registration, profile management, membership tiering, points accounting, coupon wallet, and preference tracking.
Order Management	Order creation and validation, order assignment to stores, status tracking, queue management, and order lifecycle from placement to completion.
Store Fulfillment Management	Store-level preparation workflow, recipe display, equipment integration, task queue, and completion confirmation.
Delivery Management	Rider assignment, delivery route optimization, real-time tracking, estimated time of arrival, and delivery completion recording.
Payment Management	Payment processing, transaction logging, refund handling, settlement with payment providers, and fraud detection.
Analytics & Reporting	Business intelligence dashboards, sales reports, customer behaviour analysis, campaign effectiveness measurement, and operational KPI computation.

The flow relationships connect application interfaces and services to the components they invoke: the Mobile App and WeChat Mini Program interfaces both serve as access points to the Order & Menu Service, which flows to Order Management; Store Operations Service flows to Store Fulfillment Management; Delivery Coordination Service flows to Delivery Management; and Payment Service flows to Payment Management.

2.2.4 Data Objects

Each application component manages one or more data objects. These data objects correspond directly to the entities that will be modelled in the relational schema:

Table 6: Data objects and their mapping to database tables.

Data Object	**Maps to Table(s)**	**Managed by**
Customer & Member Data	`Customers`, `MembershipAccounts`	Customer Management
Product & Menu Data	`Products`, `RecipeStandards`	Order Management
Order & Inventory Data	`DigitalOrders`, `OrderItems`, `StoreMaterials`	Order & Store Management
Delivery Data	`FulfilmentTasks`, `DeliveryPartners`	Delivery Management
Payment Data	`Payments` (via Payment Service)	Payment Management
Analytics Data	Reporting views / OLAP store	Analytics & Reporting

2.3 Technology Layer Model (TLM)

The Technology Layer describes the physical infrastructure — devices, networks, platform hosting, system software, and technology artifacts — on which the application components run. It answers the question: *what hardware, network, and software infrastructure supports the applications?*

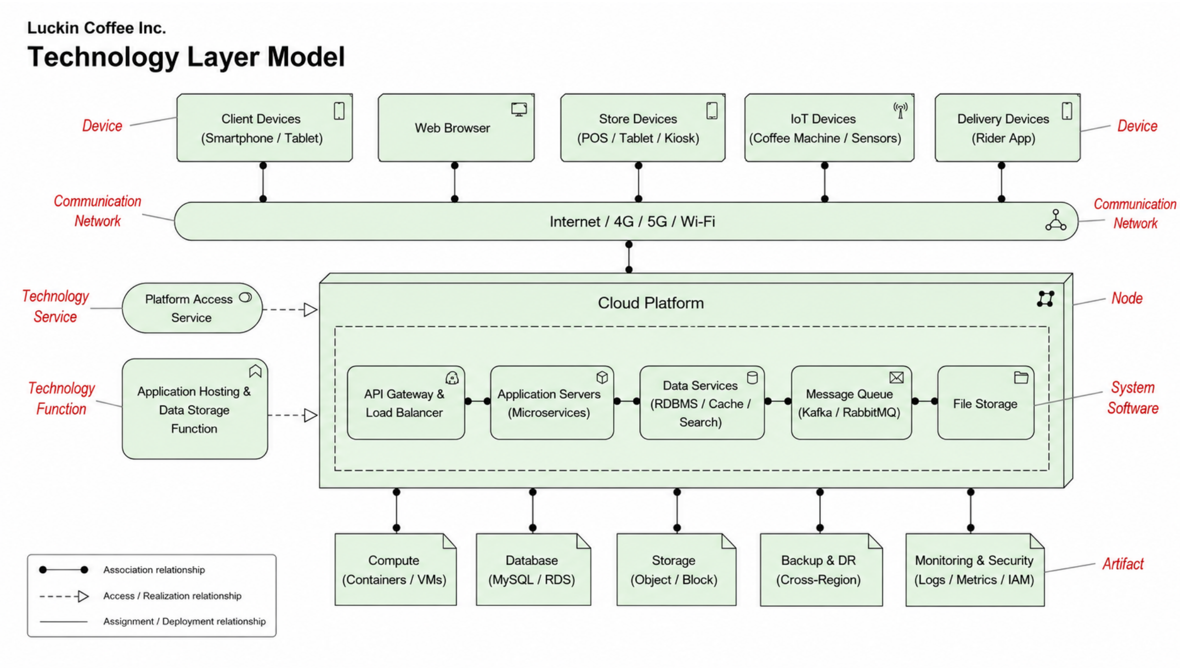


Figure 3: Technology Layer Model for Luckin Coffee — showing devices, communication networks, the cloud platform, system software, and technology artifacts.

2.3.1 Device Layer

Five categories of devices interact with the Luckin platform:

Table 7: Device types in the TLM.

Device	Type	Primary Users / Context
Client Device	Smartphone / Tablet	Customers for ordering; Store operators for fulfilment tasks.
Web Browser	Web Application	Internal staff (HQ planners, analysts, supply chain managers) for back-office functions.
Store Device	POS, Label Printer	Store staff for order receipt printing, label generation, and in-store point-of-sale operations.
IoT Device	Coffee Machine Sensors	Store equipment; sensors monitor brewing parameters, machine status, and maintenance needs.
Delivery Device	Rider App (Smartphone)	Delivery riders for receiving tasks, route navigation, and delivery confirmation.

2.3.2 Communication Network

All devices connect through a shared communication layer: **Internet / 4G / 5G / Wi-Fi**. This reflects the reality that Luckin's architecture must support diverse connectivity — customers on mobile networks, stores on broadband or Wi-Fi, IoT sensors on local wireless, and delivery riders on 4G/5G.

2.3.3 Technology Platform and System Software

The core hosting environment is a **Cloud Platform**. External traffic reaches the platform through a **Platform Access Interface** that routes requests from the communication network into the cloud infrastructure. The system software stack running on the platform comprises:

Table 8: System software components.

System Software	Function
Application Hosting & Orchestration	Container orchestration (e.g., Kubernetes), service deployment, scaling, and health monitoring.
API Gateway (Load Balancer)	Request routing, rate limiting, authentication, and load distribution across application servers.
Application Servers (Business Logic)	Runtime environment for the six application components of the Luckin Digital Platform.
Data Services (DBMS / Cache / Files)	Relational database management, in-memory caching (e.g., Redis), and file/blob storage.
Message Queue (Async Event Bus)	Asynchronous messaging for order events, payment callbacks, delivery status updates, and analytics ingestion.
File Services	Static asset serving (product images, promotional banners), log archival, and report file generation.

2.3.4 Technology Artifacts

The concrete infrastructure artifacts deployed on the cloud platform:

Table 9: Technology artifacts.

Artifact	Technology	Purpose
Compute	Containers / VMs	Runs the application servers and orchestration layer; auto-scales with demand (peak morning/lunch hours).
Database	MySQL / MongoDB	MySQL for transactional data (orders, customers, inventory); MongoDB for semi-structured analytics and log data.
Storage	Object / Block	Object storage for images, files, backups; block storage for database volumes.
Backup & DR	Cross-Region	Disaster recovery with cross-region replication; point-in-time recovery for transactional data.
Monitoring & Security	Logs / Metrics / WAF	Centralized logging, performance metrics dashboards, alerting, and Web Application Firewall for security.

3 Relational Schema Refinement and Normalization

Building on the Seminar 5 initial relational model, this section documents the normalization of the Luckin Coffee database schema to Third Normal Form (3NF), provides a data dictionary, illustrates sample data, and presents the refined relational schema diagram. The data objects identified in the Application Layer Model (Section 2) are the entities normalized here.

3.1 Normalization Process: UNF to 3NF

We demonstrate the normalization process using the core order-processing subset of the schema. The starting point is an unnormalized representation that might be produced from a naive merge of order, customer, product, and store data.

3.1.1 Unnormalized Form (UNF)

A single flat table `Customer_Orders` that contains all order-related data, including repeating product lines per order:

Table 10: UNF table — Customer_Orders (repeating groups highlighted).

Order_ID	Customer_Name	Customer_Phone	Order_Date	Store_Name, City	{Product_Name}	{Qty}
1001	Zhang Wei	138xxxx1234	2025-06-01	Store A, Beijing	Latte, Croissant	1, 2
1002	Li Na	139xxxx5678	2025-06-01	Store B, Shanghai	Americano	1

Problems: repeating groups for product lines; customer data repeated across orders; store data embedded; no atomic per-cell values in product columns.

3.1.2 First Normal Form (1NF)

Eliminate repeating groups by ensuring each cell contains exactly one atomic value. Each product in an order becomes its own row:

Table 11: 1NF table — each product line is a separate row.

OrdID	CustName	CustPhone	OrdDate	StoreName, City	ProdName	Qty	Price
1001	Zhang Wei	138xxxx1234	2025-06-01	Store A, Beijing	Latte	1	15.0
1001	Zhang Wei	138xxxx1234	2025-06-01	Store A, Beijing	Croissant	2	8.0
1002	Li Na	139xxxx5678	2025-06-01	Store B, Shanghai	Americano	1	12.0

Problem resolved: each cell is atomic. New problem: the primary key must be composite {`OrdID`, `ProdName`}, causing partial dependencies.

3.1.3 Second Normal Form (2NF)

Remove partial dependencies — non-key attributes that depend on only part of the composite primary key. `CustName` and `CustPhone` depend only on `OrdID`, not on `ProdName`. Split into `Orders` and `OrderItems`:

Table 12: 2NF tables — Orders and OrderItems separated.

Orders

OrdID	CustName	CustPhone	OrdDate
1001	Zhang Wei	138xxxx1234	2025-06-01
1002	Li Na	139xxxx5678	2025-06-01

OrderItems

OrdID[PK, FK]	ProdName	Qty	Price
1001	Latte	1	15.0
1001	Croissant	2	8.0
1002	Americano	1	12.0

3.1.4 Third Normal Form (3NF)

Remove transitive dependencies — non-key attributes that depend on other non-key attributes. In the `Orders` table, `CustPhone` depends on `CustName` (assuming one phone per customer), not directly on `OrdID`. In `OrderItems`, the composite key still contains `ProdName`, which violates 3NF: product name is not a stable identifier, and price depends on the product, not on the order line. The correct fix is to extract `ProdName` into a new `Products` table with a surrogate key `ProdID`, then replace `ProdName` in `OrderItems` with `ProdID`. Store information should also be extracted. The final 3NF decomposition:

Table 13: 3NF tables — the fully normalized schema (core subset).

Customers

CustID	CustName
1	Zhang Wei
2	Li Na

Phones

CustID$^{PK, FK}$	Phone
1	138xxxx1234
2	139xxxx5678

Stores

StoreID	StoreName
1	Store A, Beijing
2	Store B, Shanghai

Orders

OrdID	CustID^{FK}	StoreID^{FK}	OrdDate
1001	1	1	2025-06-01
1002	2	2	2025-06-01

OrderItems

OrdID$^{PK, FK}$	ProdID$^{PK, FK}$	Qty	Price
1001	101	1	15.0
1001	201	2	8.0
1002	102	1	12.0

Products

ProdID	ProdName
101	Latte
102	Americano
201	Croissant

Note on derived attributes and deliberate denormalization. The final schema in Table 19 includes attributes such as `Subtotal` (in OrderItems), `Total_Amount`, `Discount_Amount`, and `Net_Amount` (in DigitalOrders). In a strict normalization-only context, these are **derived attributes**: `Subtotal` = Quantity × Unit_Price, and `Net_Amount` = Total_Amount − Discount_Amount. Strictly speaking, derived attributes violate the non-redundancy principle of 3NF and should be computed on the fly.

However, in a high-throughput retail platform like Luckin, computing these values at query time across millions of orders would impose unacceptable latency. The schema therefore retains these fields as a **deliberate denormalization** for read performance, transaction logging (the stored value reflects the state *at the time of order*, not the current product price), and auditability. This is an explicit, well-understood trade-off between normalization purity and operational efficiency, common in production OLTP systems. The denormalized fields are protected by triggers that ensure computed values remain consistent with their source columns.

3.2 Data Dictionary

The data dictionary defines every attribute in the normalized schema with its data type, constraints, and business meaning.

Table 14: Data dictionary for the core Luckin Coffee schema.

Table / Attribute	Data Type	Constraints	Description
Customers			
Customer_ID	INT	PK, AUTO_INCREMENT	Unique identifier for each customer.
Customer_Name	VARCHAR(100)	NOT NULL	Customer's full name.
Phone	VARCHAR(20)	UNIQUE	Customer's registered phone number.
Email	VARCHAR(100)	UNIQUE, nullable	Customer's email address (optional).
Registration_Date	DATE	NOT NULL	Date the customer registered.
Member_Level	VARCHAR(20)	DEFAULT 'Regular'	Membership tier: Regular, Silver, Gold, Platinum.
MembershipAccounts			
Member_ID	INT	PK, AUTO_INCREMENT	Unique membership identifier.
Customer_ID	INT	FK → Customers, UNIQUE	One customer owns one membership account.
Points_Balance	INT	DEFAULT 0	Accumulated loyalty points.
Coupon_Value	DECIMAL(8,2)	DEFAULT 0	Total value of available coupons.
Member_Tier	VARCHAR(20)	DEFAULT 'Regular'	Membership tier.
Activity_Status	VARCHAR(20)	CHECK (IN 'Active', 'Inactive')	Current account status.
Stores			
Store_ID	INT	PK, AUTO_INCREMENT	Unique store identifier.

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Table / Attribute	Data Type	Constraints	Description
Store_Name	VARCHAR(100)	NOT NULL	Display name of the store.
Address	VARCHAR(200)	NOT NULL	Street address of the store.
City	VARCHAR(50)	NOT NULL	City where the store is located.
District	VARCHAR(50)	NOT NULL	District within the city.
Store_Type	VARCHAR(30)	CHECK (IN 'Self-operated', 'Partnership')	Ownership model of the store.
Open_Date	DATE	NOT NULL	Date the store opened for business.
Manager_ID	INT	FK → Employees, nullable	Employee ID of the store manager.
Employees			
Employee_ID	INT	PK, AUTO_INCREMENT	Unique employee identifier.
Employee_Name	VARCHAR(100)	NOT NULL	Employee's full name.
Position	VARCHAR(50)	NOT NULL	Job title (Barista, Store Manager, etc.).
Hire_Date	DATE	NOT NULL	Date the employee was hired.
Store_ID	INT	FK → Stores	Store where the employee works.
Status	VARCHAR(20)	CHECK (IN 'Active', 'Inactive')	Employment status.
Products			
Product_ID	INT	PK, AUTO_INCREMENT	Unique product identifier.
Product_Name	VARCHAR(100)	NOT NULL	Display name of the product.
Category	VARCHAR(50)	NOT NULL	Product category (Coffee, Tea, Food, etc.).
Unit_Price	DECIMAL(8,2)	NOT NULL, ≥ 0	Standard selling price.
Description	VARCHAR(500)	nullable	Product description text.

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Table / Attribute	Data Type	Constraints	Description
Status	VARCHAR(20)	CHECK (IN ‘Available’, ‘Discontinued’)	Product availability status.
Supplier_ID	INT	FK → Suppliers	Supplier of this product or its primary ingredient.
Suppliers			
Supplier_ID	INT	PK, AUTO_INCREMENT	Unique supplier identifier.
Supplier_Name	VARCHAR(100)	NOT NULL	Company name of the supplier.
Contact_Person	VARCHAR(100)	NOT NULL	Primary contact person at the supplier.
Phone	VARCHAR(20)	NOT NULL	Contact phone number.
Email	VARCHAR(100)	NOT NULL	Contact email address.
Address	VARCHAR(200)	NOT NULL	Supplier’s physical address.
Rating	INT	CHECK ($1 \leq$ Rating ≤ 5), nullable	Supplier quality rating.
DigitalOrders			
Order_ID	INT	PK, AUTO_INCREMENT	Unique order identifier.
Customer_ID	INT	FK → Customers	Customer who placed the order.
Store_ID	INT	FK → Stores	Store assigned to fulfil the order.
Order_Date	DATE	NOT NULL	Date the order was placed.
Order_Time	TIME	NOT NULL	Time the order was placed.
Total_Amount	DECIMAL(10,2)	NOT NULL, ≥ 0	Gross order amount before discounts.
Discount_Amount	DECIMAL(10,2)	DEFAULT 0, ≥ 0	Total discount applied.
Net_Amount	DECIMAL(10,2)	NOT NULL, ≥ 0	Net payable amount after discounts.

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Table / Attribute	Data Type	Constraints	Description
Order_Status	VARCHAR(30)	CHECK (IN 'Pending', 'Preparing', 'Ready', 'Delivered', 'Cancelled')	Current order status.
Order_Channel	VARCHAR(30)	CHECK (IN 'App', 'MiniProgram', 'Delivery Platform')	Channel through which order was placed.
OrderItems			
OrderItem_ID	INT	PK, AUTO_INCREMENT	Unique line-item identifier.
Order_ID	INT	FK → DigitalOrders	Parent order.
Product_ID	INT	FK → Products	Product ordered.
Quantity	INT	NOT NULL, > 0	Number of units ordered.
Unit_Price	DECIMAL(8,2)	NOT NULL, ≥ 0	Unit price at time of order.
Subtotal	DECIMAL(10,2)	NOT NULL, ≥ 0	Quantity × Unit_Price.
Payments			
Payment_ID	INT	PK, AUTO_INCREMENT	Unique payment identifier.
Order_ID	INT	FK → DigitalOrders, UNIQUE	Order this payment settles (one-to-one).
Payment_Method	VARCHAR(30)	CHECK (IN 'WeChat Pay', 'Alipay', 'Bank Card', 'Balance')	Payment method used.

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Table / Attribute	Data Type	Constraints	Description
Amount	DECIMAL(10,2)	NOT NULL, $\geq$ 0	Amount paid.
Payment_Date	DATETIME	NOT NULL	Timestamp of payment.
Transaction_Status	VARCHAR(30)	CHECK (IN 'Success', 'Failed', 'Refunded')	Payment transaction outcome.
Campaigns			
Campaign_ID	INT	PK, AUTO_INCREMENT	Unique campaign identifier.
Campaign_Name	VARCHAR(100)	NOT NULL	Marketing campaign name.
Start_Date	DATE	NOT NULL	Campaign start date.
End_Date	DATE	NOT NULL, $\geq$ Start_Date	Campaign end date.
Budget	DECIMAL(12,2)	NOT NULL, $\geq$ 0	Total campaign budget.
Status	VARCHAR(20)	CHECK (IN 'Draft', 'Active', 'Ended')	Campaign lifecycle status.
Created_By	INT	FK $\rightarrow$ Employees	Employee who created the campaign.
Promotions			
Promo_ID	INT	PK, AUTO_INCREMENT	Unique promotion identifier.
Campaign_ID	INT	FK $\rightarrow$ Campaigns	Parent campaign.
Promo_Name	VARCHAR(100)	NOT NULL	Promotion name.
Discount_Type	VARCHAR(30)	CHECK (IN 'Percentage', 'Fixed', 'BOGO')	Type of discount.

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Table / Attribute	Data Type	Constraints	Description
Discount_Value	DECIMAL(8,2)	NOT NULL, ≥ 0	Discount amount or percentage.
Start_Date	DATE	NOT NULL	Promotion start date.
End_Date	DATE	NOT NULL, $\geq$ Start_Date	Promotion end date.
Conditions	VARCHAR(300)	nullable	Conditions for the promotion to apply.
Inventory			
Inventory_ID	INT	PK, AUTO_INCREMENT	Unique inventory record identifier.
Store_ID	INT	FK $\rightarrow$ Stores	Store holding the inventory.
Product_ID	INT	FK $\rightarrow$ Products	Product the inventory tracks.
Quantity	INT	NOT NULL, ≥ 0	Current stock level.
Last_Restock_Date	DATE	nullable	Date of most recent restock.
Threshold	INT	NOT NULL, ≥ 0	Minimum stock level before triggering replenishment.
Feedback			
Feedback_ID	INT	PK, AUTO_INCREMENT	Unique feedback identifier.
Customer_ID	INT	FK $\rightarrow$ Customers	Customer submitting feedback.
Store_ID	INT	FK $\rightarrow$ Stores, nullable	Store the feedback relates to.
Rating	INT	CHECK ($1 \leq$ Rating ≤ 5)	Customer satisfaction rating.
Comment	VARCHAR(1000)	nullable	Free-text feedback comment.
Feedback_Date	DATETIME	NOT NULL	Timestamp of feedback submission.

3.3 Sample Data

The following tables show representative sample data illustrating how the normalized schema would be populated for Luckin Coffee's daily operations.

3.3.1 Customers and Membership

Table 15: Sample data — Customers.

Customer_ID	Customer_Name	Phone	Email	Registration_Date	Member_Level
1	Zhang Wei	138xxxx1234	zhangwei@email.com	2024-03-15	Gold
2	Li Na	139xxxx5678	lina@email.com	2024-06-01	Silver
3	Wang Lei	137xxxx9012	wanglei@email.com	2025-01-10	Regular
4	Chen Yue	136xxxx3456	chenyue@email.com	2025-04-22	Gold

3.3.2 Stores

Table 16: Sample data — Stores.

Store_ID	Store_Name	Address	City	Store_Type	Open_Date
1	Luckin Beijing CBD	88 Jianguo Rd, Chaoyang	Beijing	Self-operated	2023-06-01
2	Luckin Shanghai Lujiazui	168 Lujiazui Rd, Pudong	Shanghai	Self-operated	2023-08-15
3	Luckin Shenzhen Nanshan	256 Shennan Blvd, Nanshan	Shenzhen	Partnership	2024-02-20

3.3.3 Products

Table 17: Sample data — Products.

Product_ID	Product_Name	Category	Unit_Price	Status	Supplier_ID
101	Classic Americano	Coffee	12.00	Available	1
102	Iced Latte	Coffee	15.00	Available	1
103	Coconut Latte	Coffee	18.00	Available	2
201	Butter Croissant	Food	8.00	Available	3
202	Blueberry Muffin	Food	10.00	Available	3

3.3.4 Orders and Order Items

Table 18: Sample data — DigitalOrders.

Order_ID	Customer_ID^{FK}	Store_ID^{FK}	Order_Date	Total_Amount	Discount	Order_Channel
1001	1	1	2025-06-01	33.00	3.00	App
1002	2	2	2025-06-01	12.00	0.00	MiniProgram
1003	3	1	2025-06-02	46.00	5.00	App

Table 19: Sample data — OrderItems.

OrderItem_ID	Order_ID^{FK}	Product_ID^{FK}	Quantity	Unit_Price	Subtotal
1	1001	102	1	15.00	15.00
2	1001	201	2	8.00	16.00
3	1002	101	1	12.00	12.00
4	1003	103	2	18.00	36.00
5	1003	202	1	10.00	10.00

3.4 Refined Relational Schema

The complete normalized relational schema, as derived from the ER diagrams and the normalization process above, is shown in the following relation table. This refines the Seminar 5 schema by adding the new entities identified in the ArchiMate models (Campaigns, Promotions, Inventory, Feedback) and ensuring 3NF compliance.

Table 20: Refined relational data model (complete).

Relation	Attributes
Customers	Customer_ID, Customer_Name, Phone, Email, Registration_Date, Member_Level
MembershipAccounts	Member_ID, Customer_ID^{FK} (UNIQUE), Points_Balance, Coupon_Value, Member_Tier, Activity_Status
DigitalChannels	Channel_ID, Channel_Type, Access_Device, App_Version
Campaigns	Campaign_ID, Campaign_Name, Start_Date, End_Date, Budget, Status, Created_By^{FK} (Employees)
Promotions	Promo_ID, Campaign_ID^{FK}, Promo_Name, Discount_Type, Discount_Value, Start_Date, End_Date, Conditions
MemberPromotions	Member_ID$^{PK, FK}$, Promo_ID$^{PK, FK}$, Received_Date, Used_Status

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Relation	Attributes
DigitalOrders	Order_ID, Customer_ID$^{\text{FK}}$, Store_ID$^{\text{FK}}$, Channel_ID$^{\text{FK}}$, Order_Date, Order_Time, Total_Amount, Discount_Amount, Net_Amount, Order_Status, Order_Channel
OrderItems	OrderItem_ID, Order_ID$^{\text{FK}}$, Product_ID$^{\text{FK}}$, Quantity, Unit_Price, Subtotal
Payments	Payment_ID, Order_ID$^{\text{FK}}$ (UNIQUE), Payment_Method, Amount, Payment_Date, Transaction_Status
Stores	Store_ID, Store_Name, Address, City, District, Store_Type, Open_Date, Manager_ID$^{\text{FK}}$ (Employees, nullable)
Employees	Employee_ID, Employee_Name, Position, Hire_Date, Store_ID$^{\text{FK}}$, Status
Products	Product_ID, Product_Name, Category, Unit_Price, Description, Status, Supplier_ID$^{\text{FK}}$
Suppliers	Supplier_ID, Supplier_Name, Contact_Person, Phone, Email, Address, Rating
Inventory	Inventory_ID, Store_ID$^{\text{FK}}$, Product_ID$^{\text{FK}}$, Quantity, Last_Restock_Date, Threshold
Feedback	Feedback_ID, Customer_ID$^{\text{FK}}$, Store_ID$^{\text{FK}}$ (nullable), Rating, Comment, Feedback_Date
FulfilmentTasks	Task_ID, Order_ID$^{\text{FK}}$, Store_ID$^{\text{FK}}$, Task_Type, Prep_Time, Waiting_Time, Completion_Rate, Status
DeliveryPartners	Partner_ID, Partner_Name, Partner_Type, Service_Area, Standard_Delivery_Fee, Complaint_Rate
RecipeStandards	Recipe_ID, Product_ID$^{\text{FK}}$, Recipe_Version, Sop_Status, Recipe_Standard_Cost, Defect_Rate, Effective_From, Effective_To
RawMaterials	Material_ID, Supplier_ID$^{\text{FK}}$, Material_Name, Material_Type, Origin, Batch_Status, Inventory_Value
StoreMaterials	Store_ID$^{\text{PK, FK}}$, Material_ID$^{\text{PK, FK}}$, Stock_Level, Stockout_Rate
RecipeMaterials	Recipe_ID$^{\text{PK, FK}}$, Material_ID$^{\text{PK, FK}}$, Dosage_Amount, Dosage_Unit

3.5 Foreign-Key Dependencies

Table 21: Main foreign-key dependencies in the refined schema.

Foreign Key in Relation	References	Meaning
MembershipAccounts.Customer_ID	Customers.Customer_ID	One customer owns one membership account (UNIQUE constraint enforces one-to-one).
DigitalOrders.Customer_ID	Customers.Customer_ID	A customer can place many orders.
DigitalOrders.Store_ID	Stores.Store_ID	Each order is fulfilled by one store.
DigitalOrders.Channel_ID	DigitalChannels.Channel_ID	Each order is placed through one channel.
OrderItems.Order_ID	DigitalOrders.Order_ID	One order can contain multiple line items.
OrderItems.Product_ID	Products.Product_ID	Each line item references one product.
Payments.Order_ID	DigitalOrders.Order_ID	Each order has at most one payment (UNIQUE constraint enforces one-to-one).
Promotions.Campaign_ID	Campaigns.Campaign_ID	One campaign can have multiple promotions.
Campaigns.Created_By	Employees.Employee_ID	Each campaign is created by one employee.
Employees.Store_ID	Stores.Store_ID	An employee works at one store.
Stores.Manager_ID	Employees.Employee_ID	A store is managed by one employee (nullable).
Products.Supplier_ID	Suppliers.Supplier_ID	Each product is supplied by one supplier.
Inventory.Store_ID	Stores.Store_ID	Inventory tracked per store.
Inventory.Product_ID	Products.Product_ID	Inventory tracked per product.
Feedback.Customer_ID	Customers.Customer_ID	A customer can submit multiple feedback records.
Feedback.Store_ID	Stores.Store_ID	Feedback may relate to a specific store (nullable).
MemberPromotions.Member_ID	MembershipAccounts.Member_ID	Bridge: member receives promotion.
MemberPromotions.Promo_ID	Promotions.Promo_ID	Bridge: promotion issued to member.
FulfilmentTasks.Order_ID	DigitalOrders.Order_ID	Task traces back to an order.
FulfilmentTasks.Store_ID	Stores.Store_ID	Task is executed at a store.
RawMaterials.Supplier_ID	Suppliers.Supplier_ID	Material supplied by a supplier.
StoreMaterials.Store_ID	Stores.Store_ID	Bridge: store stocks material.
StoreMaterials.Material_ID	RawMaterials.Material_ID	Bridge: material stocked at store.
RecipeStandards.Product_ID	Products.Product_ID	Product has recipe versions.
RecipeMaterials.Recipe_ID	RecipeStandards.Recipe_ID	Bridge: recipe uses material.
RecipeMaterials.Material_ID	RawMaterials.Material_ID	Bridge: material used in recipe.

3.6 Integrity Rules

Table 22: Examples of integrity rules implemented in the refined schema.

Rule Type	Example in the Luckin Coffee Schema
Entity Integrity	Every base relation has a defined primary key: Customer_ID, Order_ID, Product_ID, Store_ID, etc.
Referential Integrity	OrderItems.Product_ID must reference an existing Products.Product_ID; DigitalOrders.Customer_ID must reference an existing Customers.Customer_ID.
One-to-One Control	MembershipAccounts.Customer_ID is UNIQUE, ensuring one membership account per customer.
Composite Key Bridges	MemberPromotions, StoreMaterials, and RecipeMaterials use composite primary keys to resolve many-to-many relationships.
Domain Validity	Order_Status is constrained to {Pending, Preparing, Ready, Delivered, Cancelled}; Member_Level is constrained to {Regular, Silver, Gold, Platinum}; Rating is constrained to integers 1–5.
Business Arithmetic	Net_Amount = Total_Amount - Discount_Amount; Subtotal = Quantity $\times$ Unit_Price; both must be non-negative.
Time Validity	Campaigns.End_Date $\geq$ Start_Date; Promotions.End_Date $\geq$ Start_Date.
Cross-Table Rules	A promotion applied to an order must be valid for the order date; the discount applied cannot exceed the order's total amount.
Stock Consistency	Inventory.Quantity ≥ 0 at all times (no negative stock); replenishment triggered when Quantity $<$ Threshold.
Coupon–Promotion Consistency	MembershipAccounts.Coupon_Value is a summary of coupon assets whose individual status is tracked in the MemberPromotions bridge table. In a distributed or high-concurrency environment, these two sources of truth can diverge (e.g., a coupon is redeemed in MemberPromotions but Coupon_Value is not decremented). The schema enforces consistency via application-level transactions that atomically update both tables, and a periodic reconciliation job flags mismatches between Coupon_Value and the sum of unused coupon records in MemberPromotions.

4 Business Scenarios Supported by the Refined Schema

The refined schema supports all business scenarios documented in the Seminar 5 report, and adds new scenarios enabled by the Campaigns, Promotions, Inventory, and Feedback tables introduced in the ArchiMate models.

4.1 Scenario 1: Mobile Order and Pickup with Promotions

When a customer opens the app, browses available promotions, places a coffee order with a coupon applied, and picks it up:

- **Relations joined**: Customers → MembershipAccounts → MemberPromotions → Promotions → Campaigns → DigitalOrders → OrderItems → Products → Payments → FulfilmentTasks → Stores.
- **Business value**: The schema traces the full promotional journey — which campaign issued the coupon, whether the member used it, which products it applied to, and whether fulfilment was timely. This enables campaign ROI analysis.

4.2 Scenario 2: Store-Level Inventory Replenishment

When a store's inventory of a product or material falls below its threshold:

- **Relations joined**: Stores → Inventory → Products; Stores → StoreMaterials → RawMaterials → Suppliers.
- **Business value**: The system can identify low-stock items by store, group replenishment needs by supplier, and trigger automatic purchase orders. Inventory data also feeds demand forecasting models.

4.3 Scenario 3: Customer Feedback and Service Improvement

When a customer submits a rating and comment after order completion:

- **Relations joined**: Feedback → Customers → DigitalOrders → Stores → FulfilmentTasks.
- **Business value**: Low ratings can be correlated with specific stores, order types, fulfilment delays, or product quality issues. This enables data-driven service improvement at the store and process level.

4.4 Scenario 4: Campaign Performance Analysis

When the marketing team evaluates a campaign's effectiveness:

- **Relations joined**: Campaigns → Promotions → MemberPromotions → MembershipAccounts → DigitalOrders → OrderItems → Products.
- **Business value**: Analysts can compute redemption rates, incremental revenue per campaign, and customer segment responsiveness, enabling data-driven marketing budget allocation.

5 Alignment Between Architecture Layers and Database Schema

A key deliverable of Seminar 6 is demonstrating how the ArchiMate layered architecture models map to the relational database design. The following table traces key entities from business concept through application data object to database table:

Table 23: Vertical traceability: business concept → application data → database table.

Business Concept (BLM)	**Data Object (ALM)**	**Database Table**	**How the Table Realizes the Concept**
Customer (Business Actor)	Customer & Member Data	Customers, MembershipAccounts	Stores customer identity, contact details, membership tier, and points balance.
Fulfill Customer Order (Business Process)	Order & Inventory Data	DigitalOrders, OrderItems, FulfilmentTasks	Captures every order, its line items, and the fulfilment workflow from preparation to delivery.
Coffee Retail Service (Business Service)	Product & Menu Data	Products, RecipeStandards, RecipeMaterials	Defines the product catalogue, standard recipes, and ingredient composition.
Supplier (Business Partner)	(via Product Data)	Suppliers, RawMaterials	Records supplier identity, contact, rating, and the materials they provide.
Store Operator (Business Role)	(via Order Data)	Stores, Employees, Inventory	Represents store locations, staff assignments, and local inventory levels.
Settle Payment (Subprocess)	Payment Data	Payments	Records every payment transaction, method, and status for each order.
Customer Feedback (implied)	Customer & Member Data	Feedback	Captures post-order ratings and comments linked to customers and stores.
Campaigns & Promotions (implied)	Customer & Member Data	Campaigns, Promotions, MemberPromotions	Manages marketing campaigns, discount promotions, issuance to members, and redemption tracking.

This alignment ensures that the database schema is not an isolated technical artifact — it is the persistent data layer that supports every business process described in the Business Layer

Model and every application component described in the Application Layer Model.

6 Conclusion

This report completed the Seminar 6 tasks for the Luckin Coffee case study in two parts.

First, it presented three ArchiMate layered architecture diagrams — Business Layer Model (BLM), Application Layer Model (ALM), and Technology Layer Model (TLM) — that describe the enterprise from stakeholder-facing processes down to cloud infrastructure. The BLM identified the core business actors (Customer, Store Operator, Supplier), the Coffee Retail Service, and the Fulfill Customer Order process. The ALM mapped these to six application services, the Luckin Digital Platform with six internal components, and six managed data objects. The TLM detailed the device ecosystem, communication network, cloud platform, system software stack, and technology artifacts that host the platform. The three layers are vertically integrated, ensuring traceability from business requirement to infrastructure deployment.

Second, the report refined the relational database schema from Seminar 5. It documented the normalization process from UNF to 3NF, produced a comprehensive data dictionary defining every attribute with types and constraints, provided representative sample data, and presented the complete refined schema with 22 relations. The schema now includes new entities (Campaigns, Promotions, Inventory, Feedback) identified through the ArchiMate modelling process, and enforces integrity rules covering entity integrity, referential integrity, domain validity, business arithmetic, and time validity.

The alignment section demonstrated how every business concept in the BLM traces through an application data object in the ALM to a concrete database table, confirming that the schema fully supports the business architecture. This integrated approach — ArchiMate models for architecture description and a normalized relational schema for data persistence — provides a complete foundation for implementing the Luckin Coffee digital platform.

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